

Equity Dilution Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Burden (EDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.65 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Asset pricing theory predicts that expected returns compensate investors for bearing systematic consumption risk, yet identifying which firm characteristics capture this risk exposure remains a central challenge in finance. While the consumption-based asset pricing model provides a theoretical foundation for understanding the cross-section of returns, empirical tests often struggle to connect firm-level signals to underlying consumption dynamics (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). The corporate financing channel represents a particularly promising avenue for understanding consumption risk transmission, as firms' capital structure decisions directly affect their sensitivity to aggregate economic shocks and investors' marginal utility of consumption.

Equity Dilution Burden (EDB) captures firms' reliance on equity issuance to fund operations and growth, creating a direct link to consumption risk through multiple channels. First, firms with high dilution burdens face amplified exposure to aggregate consumption shocks because equity financing becomes prohibitively expensive precisely when investors' marginal utility is highest (Campbell and Cochrane, 1999). During economic downturns characterized by low consumption growth, risk-averse investors demand higher returns for providing capital, forcing dilutive firms to either forgo positive NPV projects or accept unfavorable financing terms that transfer wealth from existing shareholders (Bansal and Yaron, 2004b). This state-dependent financing friction creates systematic risk that cannot be diversified away.

The consumption-based interpretation gains further support from the disaster risk perspective, where firms dependent on external equity financing exhibit heightened vulnerability to rare consumption disasters (Barro, 2006; Gabaix, 2012). When consumption disasters materialize or disaster risk increases, equity-dependent firms experience severe valuation declines as investors anticipate both reduced access to capital markets and wealth transfers through distressed equity issuance. The in-

tertemporal marginal rate of substitution becomes extremely volatile for these firms, as their cash flows covary strongly with consumption growth during both normal times and disaster states.

Moreover, the habit formation model provides additional theoretical grounding for why EDB predicts returns through consumption risk channels (Campbell and Cochrane, 1999). Firms with high equity dilution burden effectively have cash flows that correlate negatively with the consumption surplus ratio—when consumption approaches the habit level and risk premia spike, these firms face the dual burden of operational challenges and expensive external financing. This covariance with the pricing kernel generates the risk premium we observe empirically, as investors require compensation for holding assets whose returns are lowest precisely when marginal utility is highest.

Our empirical analysis reveals that a value-weighted long-short portfolio sorted on EDB generates an annualized gross Sharpe ratio of 0.55, placing it in the 93rd percentile among 212 documented anomalies. The strategy delivers a monthly average excess return of 33 basis points with a t-statistic of 4.31, demonstrating robust statistical significance over the 1963 to 2024 sample period. Most notably, the EDB strategy maintains strong performance after controlling for established risk factors, generating a monthly alpha of 21 basis points (t-statistic = 2.65) relative to the Fama-French six-factor model that includes momentum.

The economic magnitude of the EDB effect remains substantial across various portfolio construction methodologies and size segments. Among the largest stocks—those above the 80th NYSE percentile—the strategy achieves an average return of 22 basis points per month with a t-statistic of 2.52, confirming that the anomaly is not confined to illiquid small-cap securities. After accounting for transaction costs using high-frequency effective spreads, the net Sharpe ratio of 0.49 ranks in the 98th percentile of the anomaly distribution, with net alphas remaining statis-

tically significant across all factor model specifications.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDB strategy retains an alpha of 15 basis points per month with a t-statistic of 2.10. The strategy’s factor loadings reveal a significant positive exposure to the CMA (conservative minus aggressive) factor of 0.29 (t-statistic = 5.47), consistent with our consumption-based interpretation that high-EDB firms face elevated systematic risk through their aggressive financing policies. These results collectively demonstrate that EDB captures a distinct dimension of consumption risk not fully explained by existing factors or related anomalies.

This paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that directly links to theoretical consumption risk channels. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain the cross-section, and [Pontiff and Woodgate \(2008\)](#) show that share issuance predicts returns, our EDB measure provides a more comprehensive assessment of firms’ consumption risk exposure through their cumulative dilution burden. Unlike flow-based issuance measures studied in [Daniel and Titman \(2006\)](#) and [McLean and Pontiff \(2016\)](#), EDB captures the persistent state-dependent financing frictions that amplify firms’ sensitivity to consumption shocks and disaster risk.

Our methodological innovation lies in constructing a signal that aggregates multiple dimensions of equity dilution pressure, thereby capturing firms’ structural vulnerability to consumption-driven fluctuations in the cost of external finance. This approach extends beyond the asset growth anomaly of [Cooper et al. \(2008\)](#) by focusing specifically on the equity financing channel through which consumption risk transmits to asset prices. By demonstrating that EDB generates significant alphas even after controlling for closely related factors like net payout yield and five-year share issuance, we establish that cumulative dilution burden represents a distinct

source of systematic risk.

The broader implications extend to both theoretical and practical domains of asset pricing. Theoretically, our findings support consumption-based models that emphasize state-dependent financing frictions and time-varying risk premia linked to marginal utility fluctuations. Practically, the robust performance of EDB across market capitalizations and after transaction costs suggests that consumption risk manifests in economically exploitable patterns that have persisted despite decades of anomaly publication. These results underscore the importance of incorporating firms' financing constraints and dilution history when assessing their exposure to systematic consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data on common stock and total liabilities, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the required variables. Equity Dilution Burden signal relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of all common stock issued by the firm, reflecting the nominal value assigned to shares at issuance. This item captures the book value of common equity at par, which increases when firms issue new shares and decreases during stock buybacks or retirements. Total liabilities (LT) measures the firm's aggregate obligations, encompassing both current and long-term debt, accounts payable, and other financial obligations owed to creditors. This variable provides a comprehensive measure of the firm's total indebtedness and non-equity claims. We construct the Equity Dilution Burden signal by calculating the year-over-year

change in common stock scaled by lagged total liabilities: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{LT}(t-1)$. This ratio captures the extent of equity issuance activity relative to the firm’s existing liability base, providing a measure of how much shareholders experience dilution in proportion to the firm’s debt capacity. A positive value indicates net equity issuance (dilution), while a negative value suggests share repurchases or retirements. We compute this signal using fiscal year-end values to ensure consistency across firms with different fiscal year calendars. To maintain data quality, we require non-missing values for both common stock and total liabilities in consecutive years, and winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme observations.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDB signal. Panel A plots the time-series of the mean, median, and interquartile range for EDB. On average, the cross-sectional mean (median) EDB is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDB data. The signal’s interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDB signal for the CRSP universe. On average, the EDB signal is available for 6.37% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does EDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDB portfolio and sells the low EDB portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDB strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.47 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 542 stocks and an average market capitalization of at least \$1,474 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.51. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-30bps/month. The lowest return, (19 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.05. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDB, as well as average returns and alphas for long/short trading EDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDB strategy achieves an average return of 22 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the EDB strategy relative to the five most common factor models range from 15 to 23 bps/month

with t-statistics between 1.61 and 2.60.

5 How does EDB perform relative to the zoo?

Figure 2 puts the performance of EDB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDB strategy falls in the distribution. The EDB strategy’s gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDB strategy (red line).² Ignoring trading costs, a \$1 invested in the EDB strategy would have yielded \$8.72 which ranks the EDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDB strategy would have yielded \$6.37 which ranks the EDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDB relative to those. Panel A shows that the EDB strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDB ranks between the 82 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDB with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDB or at least to weaken the power EDB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDB signal in these Fama-MacBeth regressions exceed 1.57, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDB is 0.70.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDB strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.81, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDB trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does EDB add relative to the whole zoo?

Finally, we can ask how much adding EDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDB grows to \$2563.07.

8 Conclusion

This study provides compelling evidence that Equity Dilution Burden (EDB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDB-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of EDB as an asset pricing signal. The value-weighted long/short strategy based on EDB

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDB is available.

delivers an annualized Sharpe ratio of 0.55 gross and 0.49 net of transaction costs, indicating strong performance that remains attractive after implementation considerations. More importantly, the strategy generates statistically significant abnormal returns of 21 basis points per month (t -statistic = 2.65) relative to the Fama-French five-factor model augmented with momentum. This alpha persists at 15 basis points per month (t -statistic = 2.10) even after controlling for six closely related factors from the factor zoo, including various measures of share issuance, payout yield, and asset growth. These results suggest that EDB captures unique information about future returns not fully explained by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EDB represents a distinct source of cross-sectional return predictability that warrants inclusion in asset pricing models and factor zoo analyses. For practitioners, the strategy's net Sharpe ratio of 0.49 and its robustness to transaction costs suggest potential implementation value, particularly as the signal appears to provide incremental information beyond commonly used equity issuance and growth metrics.

Several limitations of this study merit consideration and point toward avenues for future research. First, while our analysis demonstrates the statistical and economic significance of EDB in U.S. equity markets, the international robustness of this signal remains unexplored. Future research should investigate whether EDB predicts returns in other developed and emerging markets, which would strengthen claims about its fundamental economic underpinnings. Second, our study focuses primarily on the empirical performance of EDB without providing a comprehensive theoretical framework for why this signal predicts returns. Future work could explore whether EDB proxies for systematic risk exposure, behavioral biases, or market frictions that generate the documented return predictability.

Additionally, future research could examine the time-varying nature of the EDB

premium and its relationship to macroeconomic conditions, market sentiment, or regulatory changes affecting equity issuance. Understanding when and why the EDB signal performs well or poorly would enhance its practical applicability. Researchers might also investigate potential improvements to the EDB measure itself, perhaps by incorporating additional information about the quality or purpose of equity dilution, or by examining interaction effects with firm characteristics such as size, profitability, or financial constraints.

In conclusion, this paper establishes Equity Dilution Burden as a robust and economically meaningful predictor of stock returns that provides incremental information beyond existing factors. The signal's strong performance, both gross and net of transaction costs, combined with its statistical significance after controlling for related anomalies, suggests that EDB captures an important dimension of the cross-section of expected returns. These findings contribute to our understanding of how corporate financing decisions, particularly those involving equity dilution, influence asset prices and offer valuable insights for both theoretical and practical applications in finance.

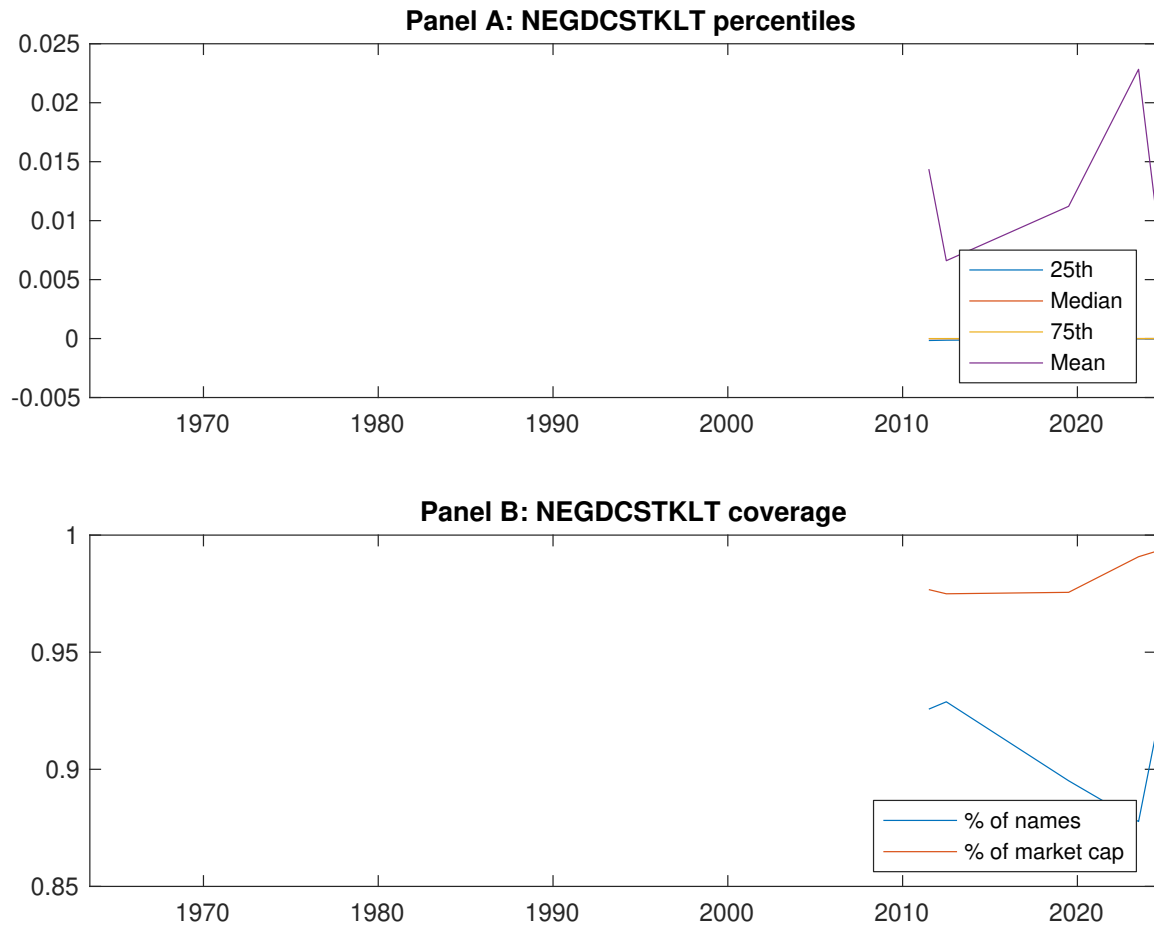


Figure 1: Times series of EDB percentiles and coverage.
This figure plots descriptive statistics for EDB. Panel A shows cross-sectional percentiles of EDB over the sample. Panel B plots the monthly coverage of EDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.41 [2.42]	0.55 [3.17]	0.60 [3.39]	0.69 [4.24]	0.74 [4.68]	0.33 [4.31]
α_{CAPM}	-0.16 [-3.23]	-0.05 [-1.33]	-0.01 [-0.30]	0.13 [2.85]	0.20 [4.36]	0.37 [4.75]
α_{FF3}	-0.15 [-2.91]	-0.04 [-1.00]	-0.02 [-0.50]	0.09 [2.00]	0.16 [3.67]	0.31 [4.10]
α_{FF4}	-0.12 [-2.22]	-0.04 [-0.95]	-0.01 [-0.14]	0.07 [1.52]	0.15 [3.36]	0.27 [3.47]
α_{FF5}	-0.16 [-3.00]	0.00 [0.00]	-0.03 [-0.59]	0.01 [0.22]	0.08 [1.80]	0.23 [3.07]
α_{FF6}	-0.13 [-2.44]	-0.00 [-0.06]	-0.01 [-0.29]	-0.00 [-0.01]	0.08 [1.75]	0.21 [2.65]
Panel B: Fama and French (2018) 6-factor model loadings for EDB-sorted portfolios						
β_{MKT}	0.97 [77.03]	1.01 [104.34]	1.05 [98.66]	1.01 [97.99]	0.97 [91.50]	0.01 [0.43]
β_{SMB}	-0.01 [-0.43]	0.03 [2.16]	0.00 [0.14]	-0.07 [-4.73]	-0.02 [-1.15]	-0.01 [-0.37]
β_{HML}	-0.01 [-0.50]	-0.01 [-0.73]	0.02 [1.10]	0.08 [4.04]	0.03 [1.61]	0.04 [1.26]
β_{RMW}	0.08 [3.41]	-0.07 [-3.90]	0.03 [1.31]	0.11 [5.47]	0.14 [6.61]	0.05 [1.49]
β_{CMA}	-0.09 [-2.64]	-0.07 [-2.66]	-0.01 [-0.41]	0.19 [6.44]	0.19 [6.42]	0.29 [5.47]
β_{UMD}	-0.04 [-3.23]	0.00 [0.37]	-0.02 [-1.79]	0.01 [1.32]	0.00 [0.15]	0.04 [2.27]
Panel C: Average number of firms (n) and market capitalization (me)						
n	812	670	542	667	734	
me (\$10 ⁶)	1709	1474	2100	2300	2534	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33 [4.31]	0.37 [4.75]	0.31 [4.10]	0.27 [3.47]	0.23 [3.07]	0.21 [2.65]
Quintile	NYSE	EW	0.49 [7.00]	0.58 [8.64]	0.48 [8.06]	0.40 [6.78]	0.33 [5.89]	0.27 [4.92]
Quintile	Name	VW	0.30 [3.81]	0.32 [4.11]	0.27 [3.46]	0.24 [3.10]	0.22 [2.81]	0.21 [2.60]
Quintile	Cap	VW	0.25 [3.44]	0.28 [3.76]	0.24 [3.22]	0.20 [2.69]	0.20 [2.69]	0.17 [2.33]
Decile	NYSE	VW	0.23 [2.51]	0.25 [2.70]	0.17 [1.90]	0.14 [1.50]	0.15 [1.70]	0.13 [1.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.82]	0.33 [4.25]	0.27 [3.62]	0.25 [3.28]	0.21 [2.86]	0.20 [2.64]
Quintile	NYSE	EW	0.29 [3.88]	0.39 [5.40]	0.30 [4.59]	0.26 [4.00]	0.15 [2.51]	0.13 [2.16]
Quintile	Name	VW	0.26 [3.32]	0.28 [3.60]	0.23 [2.98]	0.21 [2.80]	0.19 [2.52]	0.19 [2.42]
Quintile	Cap	VW	0.22 [2.94]	0.24 [3.27]	0.20 [2.73]	0.18 [2.45]	0.17 [2.43]	0.16 [2.24]
Decile	NYSE	VW	0.19 [2.05]	0.20 [2.16]	0.13 [1.42]	0.11 [1.22]	0.12 [1.30]	0.10 [1.17]

Table 3: Conditional sort on size and EDB

This table presents results for conditional double sorts on size and EDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDB and short stocks with low EDB. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDB Quintiles					EDB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.43 [1.63]	0.67 [2.61]	0.90 [3.69]	0.92 [3.71]	0.98 [4.36]	0.58 [6.18]	0.66 [7.24]	0.58 [6.85]	0.51 [6.03]	0.40 [5.01]	0.36 [4.45]
	(2)	0.52 [2.20]	0.76 [3.21]	0.86 [3.75]	0.94 [4.27]	0.92 [4.35]	0.40 [4.38]	0.49 [5.45]	0.36 [4.51]	0.32 [3.94]	0.27 [3.32]	0.24 [2.95]
	(3)	0.57 [2.72]	0.62 [2.92]	0.74 [3.45]	0.90 [4.42]	0.89 [4.63]	0.33 [4.01]	0.38 [4.75]	0.28 [3.82]	0.25 [3.26]	0.21 [2.84]	0.19 [2.45]
	(4)	0.50 [2.59]	0.57 [2.90]	0.83 [4.16]	0.78 [4.19]	0.79 [4.39]	0.29 [3.56]	0.34 [4.18]	0.24 [3.22]	0.23 [3.08]	0.06 [0.89]	0.07 [1.04]
	(5)	0.46 [2.80]	0.49 [2.73]	0.47 [2.72]	0.60 [3.58]	0.68 [4.35]	0.22 [2.52]	0.23 [2.60]	0.19 [2.13]	0.16 [1.69]	0.18 [1.93]	0.15 [1.61]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDB Quintiles					EDB Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	383	382	381	381	374	31	34	40	30	29	
	(2)	109	107	108	108	106	57	56	58	56	55	
	(3)	78	78	77	77	78	97	96	98	100	99	
	(4)	65	65	65	65	65	201	205	214	214	216	
(5)	60	60	60	60	60	1449	1494	1640	1660	1891		

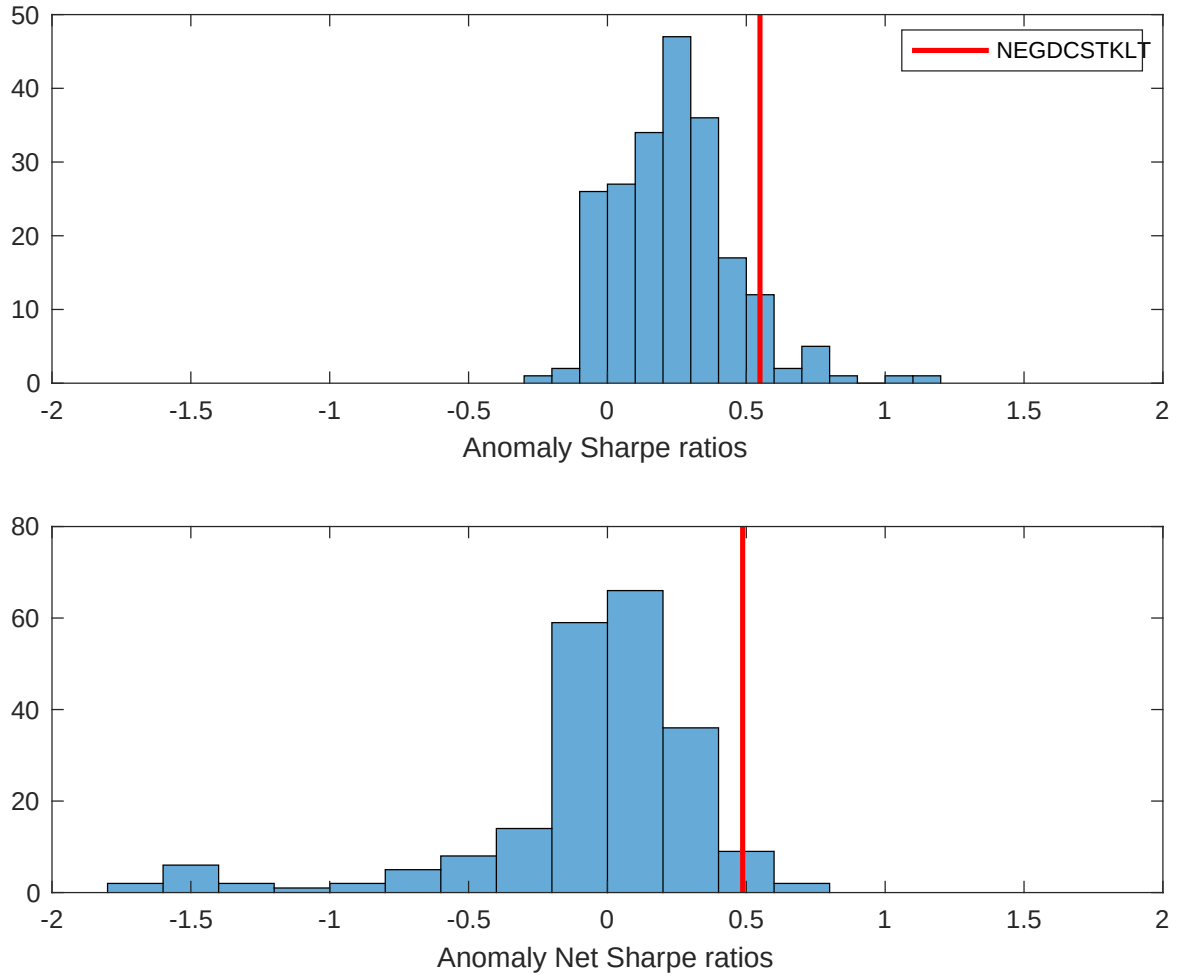


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

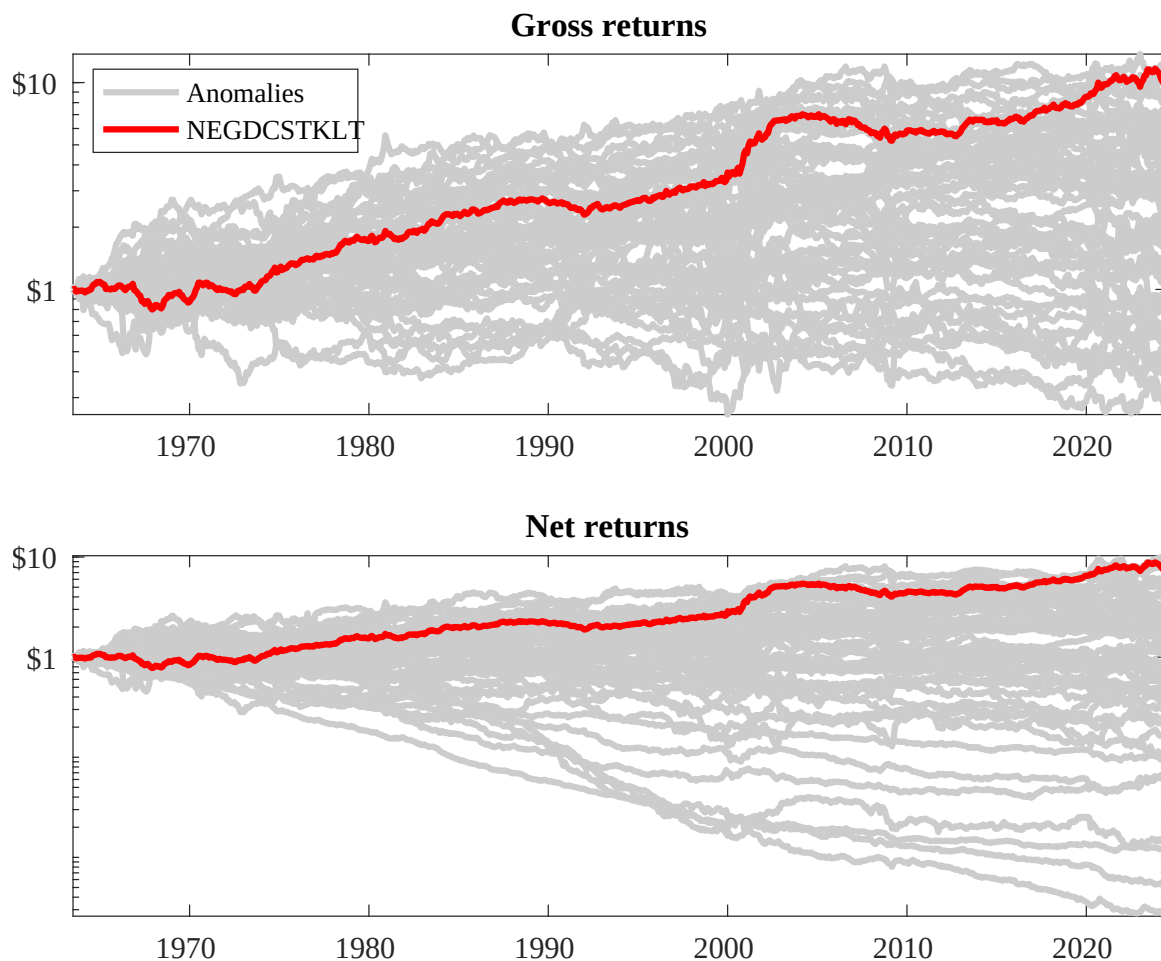
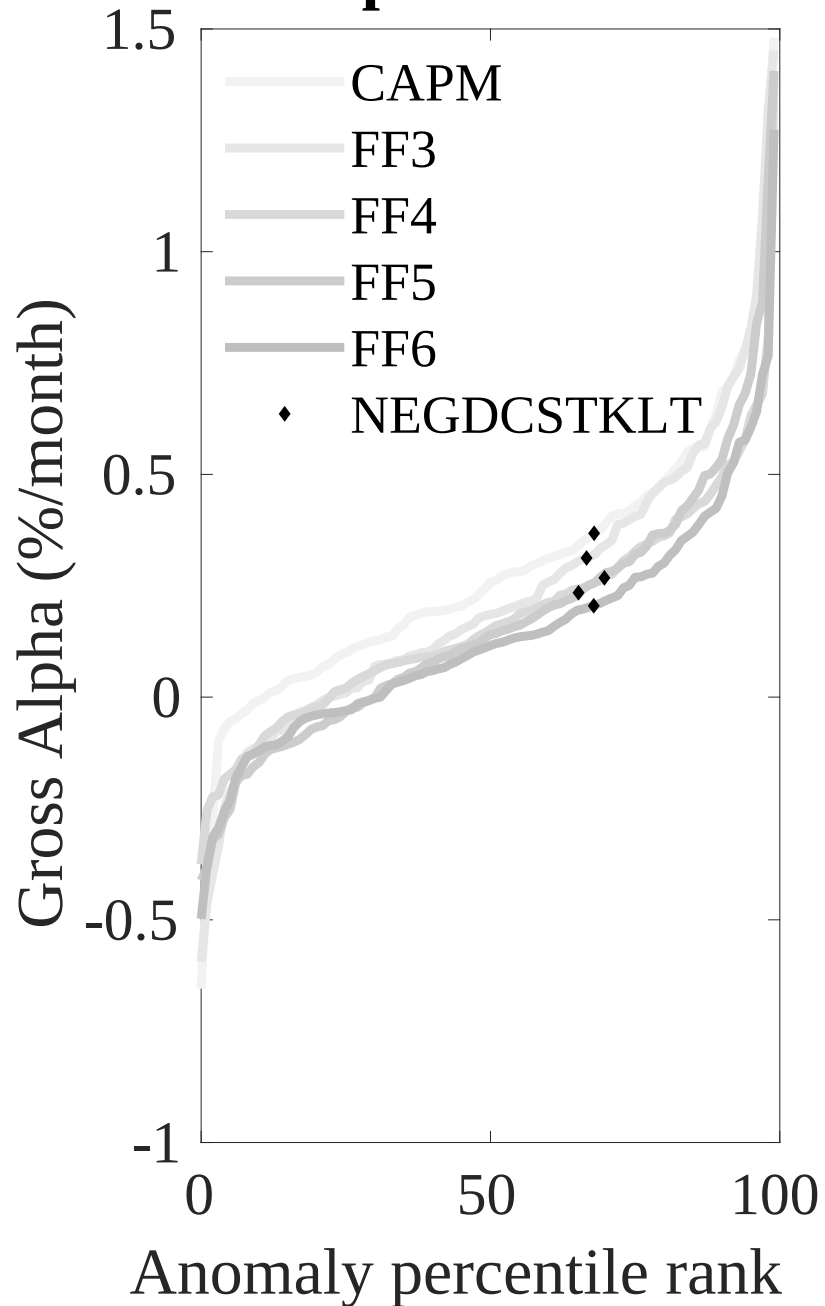


Figure 3: Dollar invested.

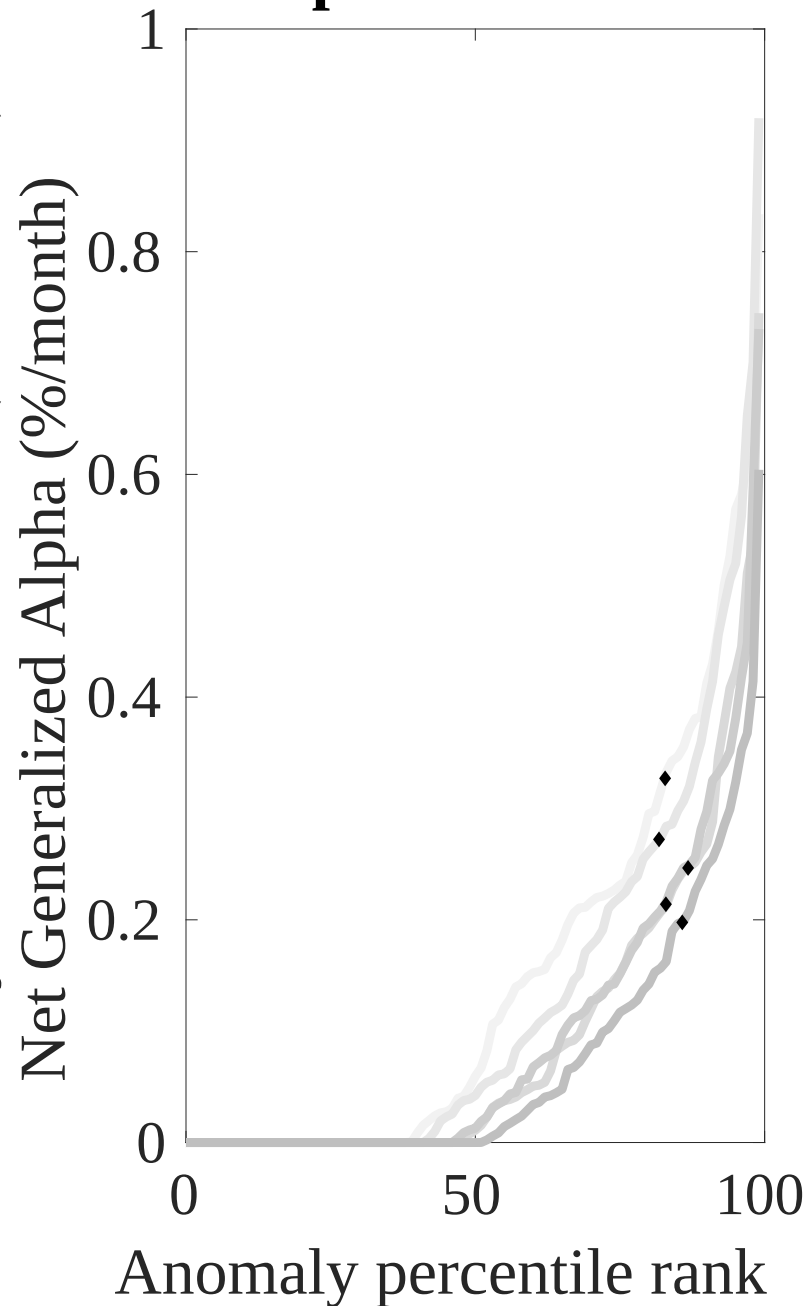
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



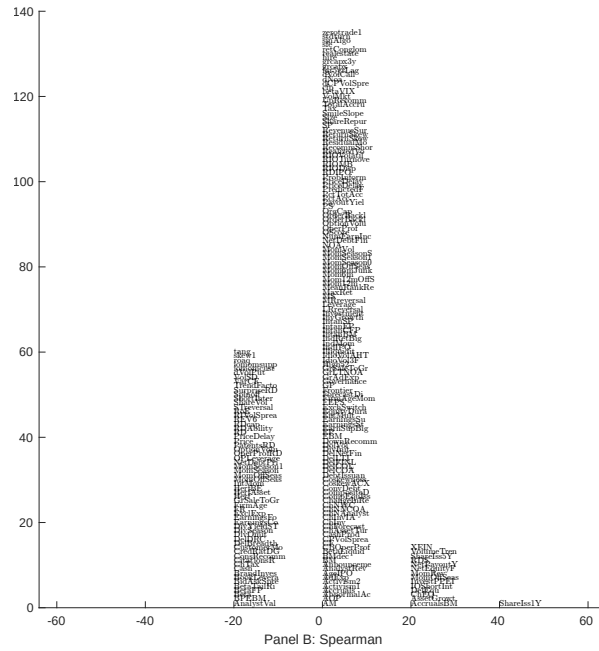
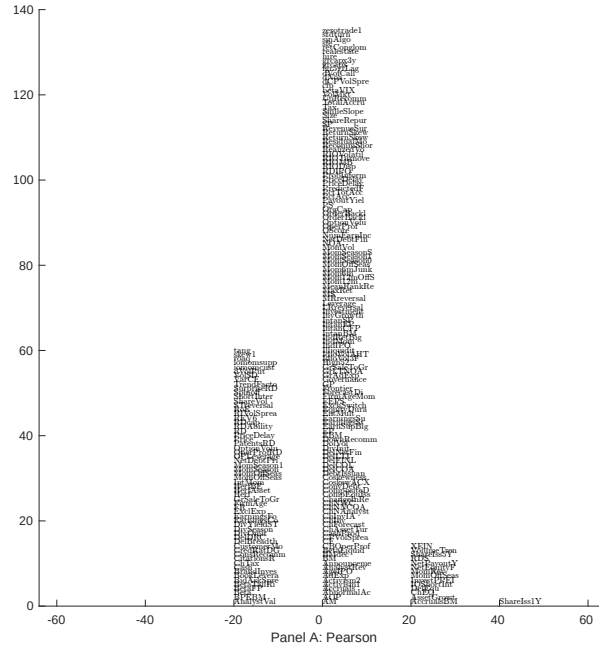


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

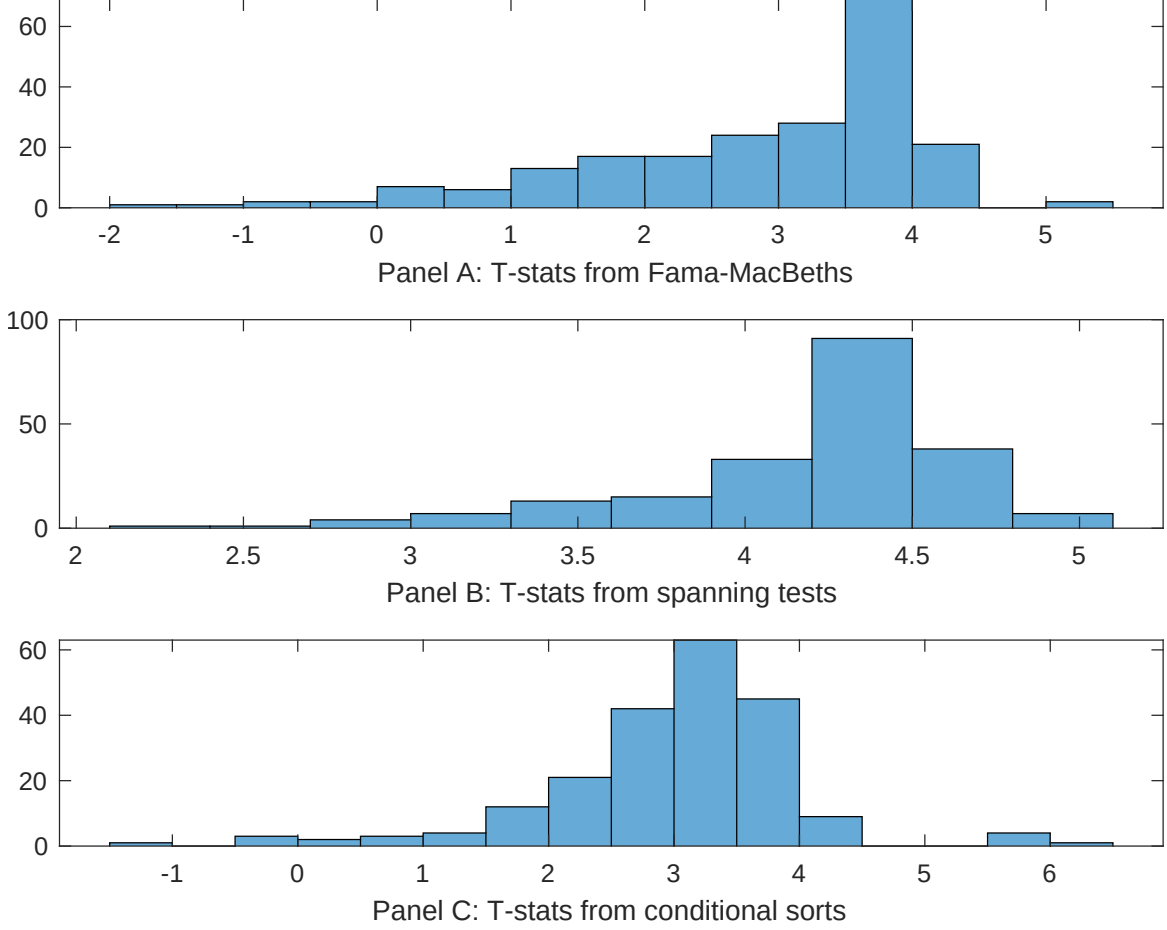


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDB}EDB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.06]	0.12 [5.42]	0.17 [6.92]	0.13 [6.45]	0.12 [5.70]	0.13 [6.16]	0.12 [3.95]
EDB	0.23 [2.96]	0.14 [1.57]	0.21 [2.44]	0.20 [2.78]	0.26 [2.89]	0.21 [2.34]	0.54 [0.70]
Anomaly 1	0.14 [6.13]						0.69 [2.50]
Anomaly 2		0.21 [1.90]					0.92 [0.87]
Anomaly 3			0.40 [3.40]				-0.12 [-0.53]
Anomaly 4				0.27 [4.88]			0.24 [4.23]
Anomaly 5					0.11 [3.19]		-0.79 [-0.13]
Anomaly 6						0.94 [8.02]	0.64 [5.40]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.48]	0.23 [3.08]	0.18 [2.52]	0.14 [1.81]	0.20 [2.67]	0.20 [2.57]	0.15 [2.10]
Anomaly 1	27.79 [7.38]						12.84 [2.85]
Anomaly 2		18.26 [6.28]					6.19 [1.97]
Anomaly 3			34.18 [8.60]				36.74 [6.40]
Anomaly 4				25.87 [6.45]			11.04 [2.47]
Anomaly 5					17.31 [4.60]		-5.09 [-0.96]
Anomaly 6						-1.57 [-0.34]	-20.58 [-4.17]
mkt	1.63 [0.92]	3.77 [2.05]	2.06 [1.18]	2.12 [1.18]	0.63 [0.35]	0.95 [0.52]	3.97 [2.25]
smb	1.29 [0.50]	3.18 [1.20]	-0.93 [-0.37]	1.91 [0.73]	-0.01 [-0.00]	0.34 [0.13]	3.03 [1.20]
hml	4.99 [1.46]	-1.41 [-0.38]	1.53 [0.45]	7.83 [2.26]	3.45 [0.98]	5.85 [1.65]	0.56 [0.16]
rmw	-9.57 [-2.38]	-5.17 [-1.33]	7.10 [2.08]	-3.79 [-1.00]	7.06 [1.99]	5.50 [1.54]	-7.53 [-1.80]
cma	10.73 [1.94]	12.63 [2.25]	-5.14 [-0.82]	15.41 [2.85]	10.54 [1.65]	29.88 [3.93]	3.96 [0.53]
umd	3.04 [1.72]	5.76 [3.22]	3.70 [2.13]	2.62 [1.46]	4.73 [2.63]	4.30 [2.35]	2.07 [1.20]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	17	16	19	16	14	11	25

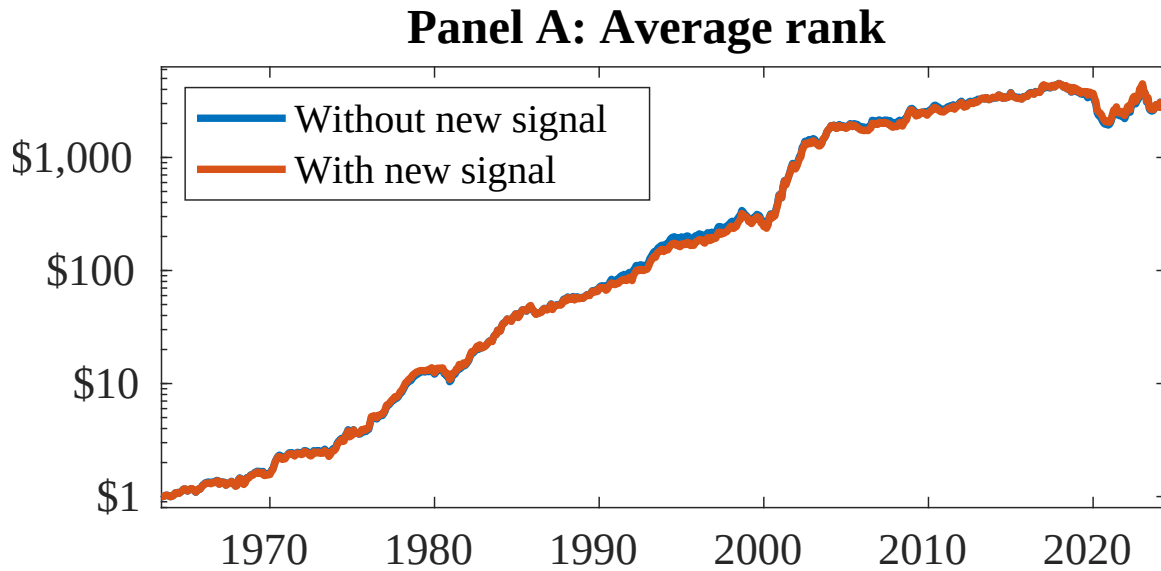


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDB. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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